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Comparative analyses of multi-criteria methods in supplier selection problem

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Abstract

The ever-growing interest in multi-criteria decision-making methods requires research that would compare them with each other as the decision-makers would need help choosing the appropriate method. It is important to highlight the differences between them, because using different methods may result in discrepancies in the obtained rankings. Additionally, it is worthwhile for the research to take into account the problems whose solution is extremely valuable. To this end, we present a comparison of five multi-criteria decision-making methods: TOPSIS, VIKOR, COMET, SPOTIS, and MARCOS in the supplier selection problem. The conducted research showed significant differences in the evaluation of the preferences of the alternatives and some discrepancies in the obtained rankings.

Keywords: MCDM; MCDA; supplier selection; comparison; rank similarity

1. Introduction

Multi-criteria decision-making methods (MCDMs) are becoming increasingly important as more and more research is done to learn more about them. These methods are important not only in everyday life where consumer decisions are often made [13, 15], but also because of their application in areas such as medicine [27, 10], banking [40, 35, 2], the choice of supplier [41, 23], or an appropriate source of renewable energy [21, 22, 6, 17].

It is important to keep extending the research in the case of multi-criteria decision-making, as each method tries to evaluate different alternatives in a different way and a small change in the operations carried out can produce com-

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pletely different results [20, 25]. Furthermore, seeing as how these methods are used in difficult areas, it is important that the decisions made are fully consistent with the expected outcome and that they are robust and rational considering the problem at hand. However, when the decision-maker has several methods presented in front of him and they present discrepancies in rankings, a problem may arise regarding the choice of the method itself.

Ranking similarity coefficients can be used to see how two rankings differ and to assess the impact of the chosen methods. Over the years, it has been tested how the different coefficients behave in order to identify the best ones, as it is important that the resulting rankings of different MCDMs can be compared to provide valuable insights and help decision-makers choose the method that suits the problem at hand [29].

In this study, five multi-criteria decision comparison methods will be tested, namely the Technique for the Order of Prioritisation by Similarity to Ideal Solution (TOPSIS), the VIseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR), the Characteristic Objects Method (COMET), the Stable Preference Ordering Towards Ideal Solution (SPOTIS) and the Measurement Alternatives and Ranking according to COmpromise Solution (MARCOS). Two similarity coefficients, the WS rank similarity coefficient, and the weighted Spearman rank similarity coefficient will be used to compare the obtained rankings.

The rest of the article is structured as follows. In Section 2, the basic concepts and steps for each selected method and ranking similarity coefficients are presented. Section 3 brings closer a study case in which the supplier selection problem is developed. Section 4 presents the results and a discussion of the rankings obtained and the ranking similarity coefficients. Section 5 provides a summary and final conclusions of the study.

2. Preliminaries

Each method performs different calculations to obtain preferences for specific alternatives. This section presents all the steps of each method for later use in other problems. The TOPSIS method is not described, as this study uses the TOPSIS ranking presented by Chun-Ho Chen [4].

2.1. The VIKOR method

The VIKOR as the multi-criteria decision-making method was first presented by Opricovic [24]. This method is often compared to the TOPSIS method, as both are highly popular and might present different results. Throughout the years, numerous extensions of the VIKOR method were proposed which might be helpful considering specific problems, eg. in the fuzzy environment [43, 37], group decision making [16], or problems where interval numbers are present [31]. The VIKOR method in its original form is executed in the following steps.

Step 1. Determine the best and the worst values of all criteria - determine best and worst values in given problem for the profit type criteria (1) and cost type criteria (2).

$$f_j^* = \max_i f_{ij},\ f_j^- = \min_i f_{ij},\ i = 1, 2, \cdots, m,\ j = 1, 2, \cdots, n \tag{1}$$

$$f_j^* = \min_i f_{ij},\ f_j^- = \max_i f_{ij},\ i = 1, 2, \cdots, m,\ j = 1, 2, \cdots, n \tag{2}$$

Step 2. Compute S_i (3) and R_i values (4) - w_j is weight vector which describes preference provided by expert

$$S_i = \sum_{i=1}^{n} w_j \left(f_j^* - f_{ij}\right) / \left(f_j^* - f_j^-\right),\ i = 1, 2, \cdots, m,\ j = 1, 2, \cdots, n \tag{3}$$

$$R_i = \max_j \left[w_j \left(f_j^* - f_{ij}\right) / \left(f_j^* - f_j^-\right)\right],\ i = 1, 2, \cdots, m,\ j = 1, 2, \cdots, n \tag{4}$$

Step 3. Compute the values of Q_i (5) - v is introduced as a weight for the strategy of the majority of criteria, whereas 1 - v is the weight of the individual regret. These strategies could be compromised by v = 0.5.

$$Q_i = v\,(S_i - S^*) / (S^- - S^*) + (1 - v)\,(R_i - R^*) / (R^- - R^*),\ i = 1, 2, \cdots, m \tag{5}$$

Step 4. Rank the alternatives - the VIKOR method provides three rankings named S, R and Q. Each of them should be ranked in ascending order. The measures of S and R are integrated into Q for a compromise solution, the base for an agreement established by mutual concessions. It is up to the decision-maker to choose a preferred solution.

2.2. The COMET

The COMET is an innovative approach in the field of decision-making presented by Sałabun in 2015 [26], which presents a method that is fully resistant to the rank reversal paradox [33, 34], which is present when new alternatives are added to the existing solution. Moreover, this method gives an expert a new way in which the alternatives might be compared, namely through characteristic objects (COs) which define the space of the problem. This method is executed in the following steps.

Step 1. Definition of the space of the problem – r number of criteria needs to be defined which will describe the alternatives used in the problem. Additionally, the set of fuzzy numbers needs to be selected for each criterion (6).

$$\begin{aligned} C_1 &= \{\tilde{C}_{11}, \tilde{C}_{12}, ..., \tilde{C}_{1c_1}\} \\ C_2 &= \{\tilde{C}_{21}, \tilde{C}_{22}, ..., \tilde{C}_{2c_2}\} \\ &... \\ C_r &= \{\tilde{C}_{r1}, \tilde{C}_{r2}, ..., \tilde{C}_{rc_r}\} \end{aligned} \tag{6}$$

where $C_1, C_2, ..., C_r$ are ordinates with fuzzy numbers for considered criteria.

Step 2. Characteristic objects creation – The Cartesian product of fuzzy number cores is used to build characteristic objects (COs). Those objects are n-dimensional space reference points that may or may not exist in real life. Those objects can represent the problem's space. The collection of all COs is obtained in this phase (7).:

$$\begin{aligned} CO_1 &= \langle C(\tilde{C}_{11}), C(\tilde{C}_{21}), ..., C(\tilde{C}_{r1})\rangle \\ CO_2 &= \langle C(\tilde{C}_{11}), C(\tilde{C}_{21}), ..., C(\tilde{C}_{r2})\rangle \\ &... \\ CO_t &= \langle C(\tilde{C}_{1c_1}), C(\tilde{C}_{2c_2}), ..., C(\tilde{C}_{rc_r})\rangle \end{aligned} \tag{7}$$

where t represents the number of characteristic objects.

Step 3. Creation of the Matrix of Expert Judgement – an expert compares the characteristic objects in the way stated in equation (8). The more preferred objects obtain a score of 1, while the least preferred receive a value of 0. Both objects receive a score of 0.5 if they are equal.

$$\alpha_{ij} = \begin{cases} 0.0, & f_{exp}(CO_i) < f_{exp}(CO_j) \\ 0.5, & f_{exp}(CO_i) = f_{exp}(CO_j) \\ 1.0, & f_{exp}(CO_i) > f_{exp}(CO_j) \end{cases} \tag{8}$$

where f_{exp} means an expert mental judgement function. In this study, however, the TOPSIS method was used instead of an expert function. This approach is presented in [19].

The expert evaluations are then put into a matrix of expert judgement (MEJ), which is presented in equation (9)

$$MEJ = \begin{pmatrix} \alpha_{11} & \alpha_{12} & ... & \alpha_{1t} \\ \alpha_{21} & \alpha_{22} & ... & \alpha_{2t} \\ ... & ... & ... & ... \\ \alpha_{t1} & \alpha_{t2} & ... & \alpha_{tt} \end{pmatrix} \tag{9}$$

where α_{ij} is the effect of comparing CO_i and CO_j.

Next, the vector of the Summed Judgements (SJ) is calculated with equation (10).

$$SJ_i = \sum_{j=1}^{t} \alpha_{ij} \tag{10}$$

Finally, the vector P is generated as an outcome, where the estimated value of preference identified for CO_i is stored.

Step 4. The Rule Base – a fuzzy rule basis must be constructed for each characteristic object. It is accomplished through a preference for a specific characteristic object, which serves as a means of defining the grade of membership in relation to specific criteria values. The rules are created by the following equation (11).

$$IF\ C\left(\tilde{C}_{1i}\right)\ AND\ C\left(\tilde{C}_{2i}\right)\ AND\ \ldots\ THEN\ P_i \tag{11}$$

Step 5. Calculating the preference – As illustrated in equation (12), the alternatives are presented as a set of crisp numbers. Each a_{ri} represents the value of the $r-th$ criteria for a particular alternative. Mamdani's fuzzy inference approach uses those numbers to determine the preference of the $i-th$ alternative, which may then be ranked.

$$A_i = \{a_{1i}, a_{2i}, \ldots, a_{ri}\} \tag{12}$$

2.3. The SPOTIS method

The SPOTIS method is another approach to the rank-reversal free method, however, it was introduced by Dezert in 2020 [9]. This method proposes a more standard approach to solving multi-criteria decision-making problems where weights need to be determined prior to the method execution. The steps for this method are presented below.

Step 1. Define the bounds of the problem - min and max bounds of classical MCDM problem must be defined to transform MCDM problem form ill-defined to well-defined.

$$\left[S_n^{\min}, S_n^{\max}\right] = [x_1, x_2] \tag{13}$$

where, n - criterion number, x1 - min bound, x2 - max bound **Step 2. Define the ideal solution point** - define vector which includes maximum or minimum from bounds for specific criterion depending on criterion type. For profit type, the max value should be taken, for cost type, min value.

$$S^{\star} = \left(S_1^{\star}, S_2^{\star}, S_3^{\star}\right) \tag{14}$$

Step 3. Compute normalized distance matrix - for each alternative ai (i= 1,. . . , M),compute its normalized distance with respect to ideal solution for each criteria cj(j= 1,. . . , N).

$$d_{ij} = \frac{\left|A_{ij} - S_j^*\right|}{\left|S_j^{\max} - S_j^{\min}\right|} \tag{15}$$

Step 4. Compute normalized averaged distance - for each criteria cj(j= 1,. . . , N) take into account its weight and calculate final preference by executing following equation.

$$\tilde{p}_j = \sum_{j=1}^{N} w_j d_{ij} \tag{16}$$

Step 5. Rank criteria by preference - values should be ranked in ascending order.

2.4. The MARCOS method

The MARCOS method by Stević in 2020 [36]. This method was initially presented for a problem of supplier selection in healthcare industries. This method tries to incorporate a new approach to solving decision problems, namely the anti-ideal and ideal solution is considered at the initial steps of solving the problem. Additionally, a new way to determine utility functions and their further aggregation is proposed. This should help maintain stability in the problems requiring a large set of alternatives and criteria. The method is executed through the following steps.

Step 1. Define problem space – The initial step requires to define set of n criteria and m alternatives to create decision matrix.

Step 2. Create extended initial matrix – Next, the extended initial matrix X should be formed by defining ideal (AI) and anti-ideal(AAI) solution.

$$X = \begin{array}{c} AII \\ A_1 \\ A_2 \\ \cdots \\ A_m \\ AI \end{array} \begin{bmatrix} x_{aa1} & x_{aa2} & \ldots & x_{aan} \\ x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \ldots & x_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ x_{m1} & x_{22} & \cdots & x_{mn} \\ x_{ai1} & x_{ai2} & \cdots & x_{ain} \end{bmatrix} \tag{17}$$

The anti-ideal solution (AAI) which is the worst alternative is defined by equation (18), whereas the ideal solution (AI) is the best alternative in the problem at hand defined by equation (19).

$$AAI = \min_i x_{ij} \quad if\, j \in B \; and \; \max_i x_{ij} \quad if\, j \in C \tag{18}$$

$$AI = \max_i x_{ij} \quad if\, j \in B \; and \; \min_i x_{ij} \quad if\, j \in C \tag{19}$$

where B is a benefit group of criteria and C is a group of cost criteria.

Step 3. Normalize extended initial matrix – After defining anti-ideal and ideal solutions, the extended initial matrix X needs to be normalized, by applying equations (20) and (21) creating normalized matrix N.

$$n_{ij} = \frac{x_{ai}}{x_{ij}} \quad if\, j \in C \tag{20}$$

$$n_{ij} = \frac{x_{ij}}{x_{ai}} \quad if\, j \in B \tag{21}$$

Step 4. Calculate weighted normalized decision matrix – The weight for each criterion needs to be defined to present its importance in accordance to others. The weighted matrix V needs to be calculated by multiplying the normalized matrix N with the weight vector through equation (22).

$$v_{ij} = n_{ij} \times w_j \tag{22}$$

.**Step 5. Compute utility degree** – Next, the utility degree K of alternatives in relation to the anti-ideal and ideal solutions needs to by calculated by using equations (23) and (24)

$$K_i^- = \frac{\sum_{i=1}^{n} v_{ij}}{\sum_{i=1}^{n} v_{aai}} \tag{23}$$

$$K_i^+ = \frac{\sum_{i=1}^{n} v_{ij}}{\sum_{i=1}^{n} v_{ai}} \tag{24}$$

Step 6. Determination of the utility function – The utility function f of alternatives, which is the compromise of the observed alternative in relation to the ideal and anti-ideal solution, needs to be determined. Its done using equation (25).

$$f(K_i) = \frac{K_i^+ + K_i^-}{1 + \frac{1-f(K_i^+)}{f(K_i^+)} + \frac{1-f(K_i^-)}{f(K_i^-)}} \tag{25}$$

where $f\left(K_i^-\right)$ represents the utility function in relation to the anti-ideal solution and $f\left(K_i^+\right)$ represents the utility function in relation to the ideal solution.

Utility functions in relation to the ideal and anti-ideal solution are determined by applying equations (26) and (27).

$$f(K_i^-) = \frac{K_i^+}{K_i^+ + K_i^-} \tag{26}$$

$$f(K_i^+) = \frac{K_i^-}{K_i^+ + K_i^-} \tag{27}$$

Step 7. Rank preference order – Finally, rank alternatives accordingly to the values of the utility functions. The higher the value the better is an alternative.

2.5. Spearman's rank correlation coefficient

Spearman's rank correlation coefficient proved itself many times [42, 44, 11, 32, 5] and because of its popularity, it was further developed into weighted variant [7] which better reflects differences between rankings as the alternatives ranked best are more significant than the rest of the ranking. The weighted Spearman's coefficient is presented in equation (28).

$$r_w = 1 - \frac{6 \cdot \sum_{i=1}^{n} (x_i - y_i)^2 \left((N - x_i + 1) + (N - y_i + 1)\right)}{n \cdot (n^3 + n^2 - n - 1)} \tag{28}$$

2.6. WS rank correlation coefficient

WS rank similarity coefficient is a recently developed rank similarity coefficient [28], which in some way may resemble similarities with Spearman's weighted coefficient as it gives more significance to the ranks closer to the podium. However, this coefficient is constructed in such a way that the most important are a couple of first ranks, so in the problem where many alternatives are present, it would emphasize the differences in the podium more prominently. Moreover, this coefficient is asymmetrical, because of how it compares two rankings, so we can state which ranking is a reference.

$$WS = 1 - \sum_{i=1}^{n} \left(2^{-x_i} \frac{|x_i - y_i|}{\max\{|x_i - 1|, |x_i - N|\}}\right) \tag{29}$$

3. Study case

In this research, previously mentioned methods are compared in the case of resulting rankings. Those methods are the TOPSIS, the VIKOR, the COMET, the SPOTIS, and the MARCOS method. Each method's main aim is a bit different, which might reflect in the final assessment, which is why many kinds of research are conducted into selecting appropriate multi-criteria decision-making method [38, 39, 14].

In this research, we wanted to further the results obtained by Chun-Ho Chen [4] where he proposed a solution to a building material supplier selection problem using the TOPSIS method. The research of Chun-Ho Chen was chosen as it presents a topic that is highly important in the case of business logic [12, 18]. Moreover, many studies have been done in search of a solution for this specific problem [8, 30, 3, 1].

In his research, Chun-Ho Chen proposed seven criteria that were describing five alternatives. The criteria that were taken into consideration are as follows: $\mathbf{C_1}$ – Rate of qualified products (%), $\mathbf{C_2}$ – Product price (thousand dollars), $\mathbf{C_3}$ – Rate of Product market share (%), $\mathbf{C_4}$ – Supply capacity (kg/time), $\mathbf{C_5}$ – New product development rate (%), $\mathbf{C_6}$ – Delivery time (days), $\mathbf{C_7}$ – Delivery on time ratio (%). Those criteria were used to create the decision matrix which is presented in Table 1.

Table 1. Decision matrix composed by Chun-Ho Chen.

A_i	C_1	C_2	C_3	C_4	C_5	C_6	C_7
A_1	0.95	36	0.19	53	0.73	11	0.93
A_2	0.98	39	0.17	52	0.75	13	0.89
A_3	0.93	33	0.21	57	0.69	11	0.92
A_4	0.91	37	0.23	56	0.77	12	0.87
A_5	0.92	35	0.16	51	0.76	10	0.86

For each multi-criteria decision-making problem, the weights of criteria need to be determined. In his research, Chun-Ho Chen used a novel entropy-AHP approach to assess the weight of each criterion. The weights are presented in Table 2. Additionally, the type of each criterion is presented, whether its profit or cost.

Through the execution of the TOPSIS method, Chun-Ho Chen obtained ranking which is presented in Table 3. As it can be seen, the alternative A_1 was positioned first, A_3 second, A_4 third, A_5 fourth and A_2 last.

Table 2. Weights of criteria.

C_i	C_1	C_2	C_3	C_4	C_5	C_6	C_7
Weight	0.2415	0.1505	0.1122	0.1051	0.1071	0.0894	0.1942
Type	Profit	Cost	Profit	Profit	Profit	Cost	Profit

Table 3. Reference ranking acquired by Chun-Ho Chen.

A_i	A_1	A_2	A_3	A_4	A_5
Rank	1	5	2	3	4

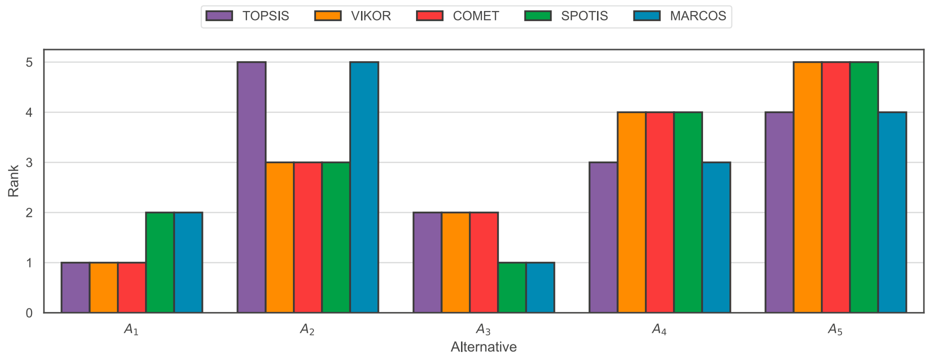

Fig. 1. Grouped bar plot of calculated rankings and reference ranking

For the remaining methods, the preferences for each alternative were calculated and are presented in Table 4. The preferences are presented because it is useful to see how the different methods evaluated the alternatives considered. In the case of the MARCOS method, very little change can be observed, namely in the second decimal place, where the differences are also not very significant. The largest scatter of preference values is shown by the VIKOR method, followed by the COMET method. It might turn out to be important to pay close attention to how different methods assesses the preference value because with a number of alternatives of several hundred or thousands the differences can be minimal which can be interpreted negatively by the decision-maker.

Table 4. Preference of alternatives calculated using different MCDMs.

A_i	VIKOR	COMET	SPOTIS	MARCOS
A_1	0.0433	0.7625	0.3963	0.6695
A_2	0.4775	0.5971	0.5614	0.6460
A_3	0.2500	0.7037	0.3692	0.6816
A_4	0.8455	0.2210	0.5854	0.6678
A_5	0.8750	0.1533	0.6821	0.6523

The resulting rankings are presented in the form of a grouped barplot, which is shown in Figure 3. Such a representation makes it easy to see the differences in the rankings obtained when few alternatives were evaluated. At first glance, we can see that the rankings are quite similar, but the biggest difference is in the case of alternative A_2, which was rated the worst by TOPSIS and MARCOS, while in the case of methods VIKOR, COMET, and SPOTIS it was placed on the third place. It is also worth noting that although the TOPSIS and SPOTIS methods similarly evaluated alternatives A_2, A_4 and A_5, in the case of alternatives A_1 and A_3 the places were reversed.

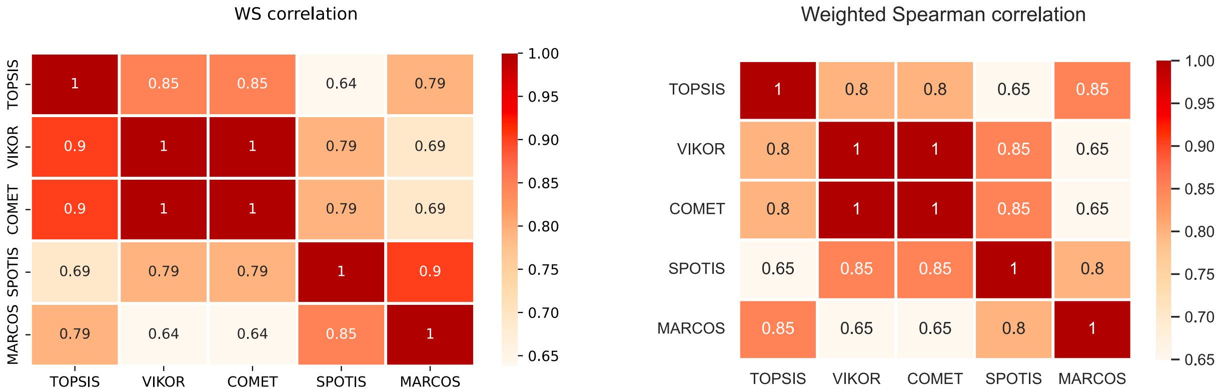

Fig. 2. (a) WS rank similarity; (b) Weighted Spearman rank similarity.

Using the obtained rankings, similarity coefficients were calculated and are presented in Figure 3. As we can read from the presented heat map, the WS coefficient and the Spearman weighted coefficient show that the COMET and VIKOR methods ranked the alternatives exactly the same. In the case of the WS coefficient, we can see that a high correlation with the reference ranking has the VIKOR and COMET methods, and then MARCOS which results from the previously described changes in the upper positions of the ranking. Additionally, quite high correlation is presented between SPOTIS and MARCOS methods. In the case of Spearman's weighted coefficient, we can see similar behavior, but the highest correlation has MARCOS method, which as we can see is caused by the fact that the three positions overlap exactly, and the weight applied to the upper positions of the ranking is smaller than in the case of the WS coefficient. In addition, quite good correlation was shown by the VIKOR and SPOTIS and COMET and SPOTIS methods.

4. Conclusions

Multi-criteria decision-making methods are designed to help the decision-maker, especially when decision-making can be extremely difficult. However, because of the great interest in the subject, many different methods have been created that calculates the preference value for each alternative in a different way which creates discrepancies in the resulting rankings. This creates a new problem in which the decision-maker may not know which method will be appropriate. Despite trying to create studies such as this one, the method should always be chosen according to the problem being solved and the expected outcome.

In this study, we presented a comparison of five methods, where we found that the VIKOR and COMET methods presented the same final ranking. Furthermore, the VIKOR, COMET, and MARCOS methods showed a high correlation with the reference ranking. The SPOTIS method showed the highest discrepancy with the reference ranking. It is likely that as the number of alternatives increases, more discrepancies could arise and thus the correlation would be lower.

For future directions, it would be useful to collect as many similar problems as possible and then carry out a comparative analysis of the multi-criteria decision-making methods to see whether the methods would behave similarly if the problem contained similar criteria, as this would allow the development of a framework for this particular problem.

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